Request for Quote: CNC-milled hardwood rotor blades, V49 geometry

Jon Farhat Inc., Burbank, California | phone: _____ email: _____ | 29 October 2026

Please quote machining and finishing of the two blade solids attached as STEP files. Quote 6 blades (3 CW + 3 CCW) and, separately, 12 blades (6 + 6). Please state unit price, any tooling or fixture charge, lead time, and shipping to Burbank, CA 91505.

1. Parts

File	Qty (6-piece order)	Qty (12-piece order)
V49_CW_BLADE_MAPLE_TE1mm.step	3	6
V49_CCW_BLADE_MAPLE_TE1mm.step (exact mirror of CW)	3	6

Models are in millimetres, single closed solids. Overall: 425.0 mm root face to tip, 35.2 mm max chord, 4.5 mm max thickness, 34.6 cm^3, about 24 g in maple. The root is a rectangular block 16.00 x 8.75 mm, 50 mm long from the root face, with two 5.0 mm through-holes on the block centreline at 9.0 and 34.0 mm from the root face (25.0 mm pitch), then a 10 mm transition into the airfoil.

2. Material

- Laminated hard maple or European beech, kiln dried, straight grain running the length of the blade (spanwise). 3 to 5 laminations preferred for stability; solid stock acceptable if quarter-sawn and defect free.
- Both blades of a CW/CCW pair cut from the same billet. No knots, checks or grain runout through the root block.

3. Tolerances

- Root block: 16.00 x 8.75 mm, +0 / -0.05 on both dimensions, flats parallel within 0.03. This block is clamped in a pitch grip; it is the reference for everything else and is a good flip-fixture datum.
- Holes: 5.00 mm +0.05 / -0, 25.0 mm centre spacing +/-0.05, first hole 9.0 +/-0.05 from the root face, on the block centreline +/-0.05, perpendicular to the 16 mm faces. Leave the holes unbushed.
- Airfoil surfaces: +/-0.20 mm to the model. Chord and twist as modeled; no added washout, no change to the leading-edge radius.
- Trailing edge: 1.0 mm thick as modeled. Do not knife-edge it. Break the trailing-edge/tip corner with a 1 mm radius.
- Overall length 425.0 +/-0.3.

4. Finish

- Machine marks sanded out (180 then 320), surface sealed with thin epoxy or an equivalent moisture seal, then clear-coated. No fill or build-up on the root block faces or in the holes; keep them bare wood.
- No logos, labels or paint colour. Light marking of CW / CCW on the root block face is welcome.
- Weight-match each CW/CCW pair within 0.5 g and report the weights.

5. Application (for your information)

Rotor blade for a two-blade rotor on a test stand. Nominal 1800 rpm, test envelope to 2500 rpm, 457 mm rotor radius, tip speed about 86 m/s nominal. The root block carries the centrifugal load (on the order of 50 lbf) through the two pins, so grain direction and the integrity of the block matter more than the airfoil tolerance.

6. Please include in your quote

- Unit price at 6 and at 12; tooling, fixture or NRE charge if any; lead time from receipt of order.
- Wood species and construction you intend to use (laminated or solid).
- Any deviation you recommend from the above, with reason. A substitute of beech for maple, or of 0.8 mm for the 1.0 mm trailing edge, is acceptable if you say so up front.
- Photos of the first finished pair before the balance of the order is finished.